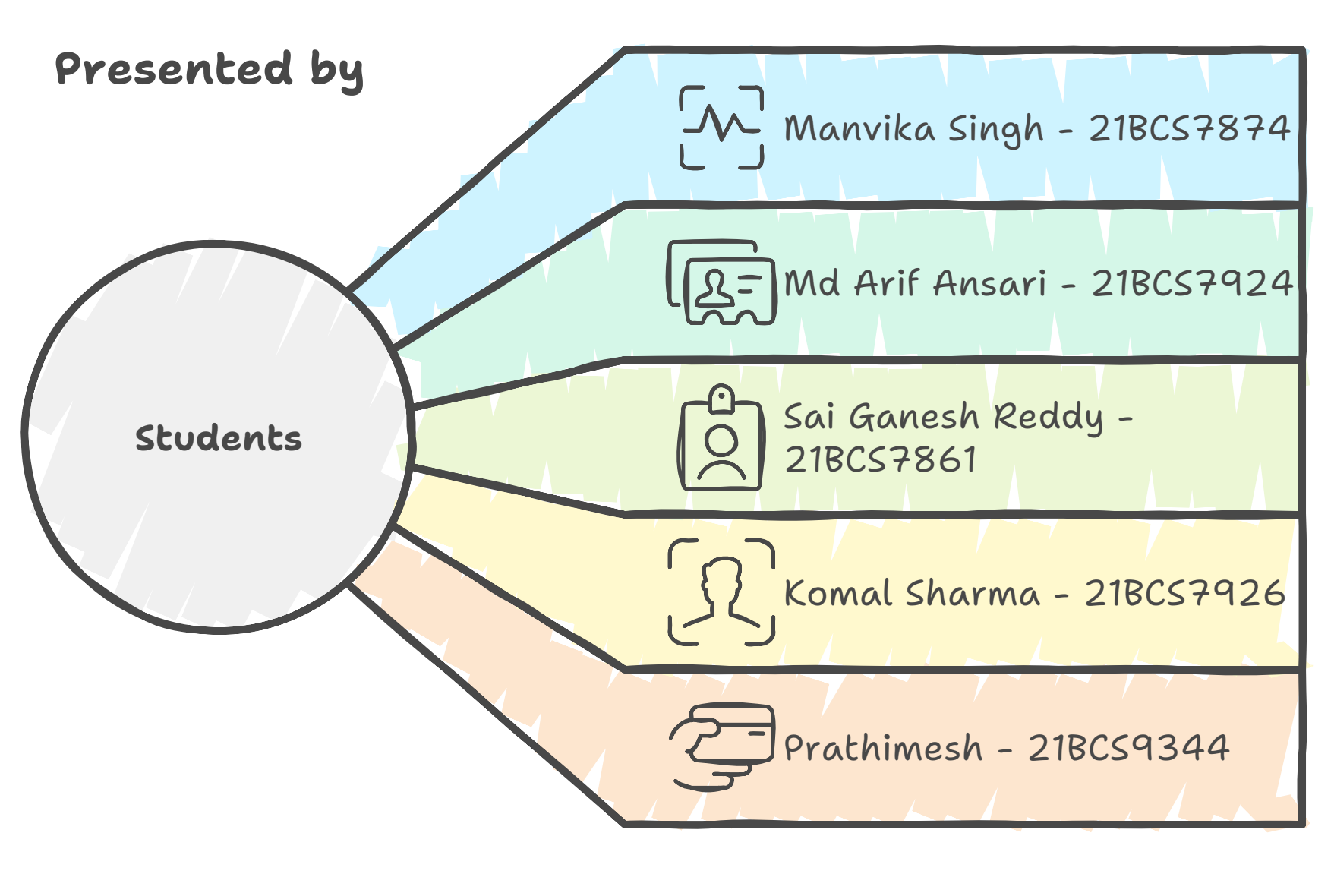
# Energy Consumption Prediction and Optimization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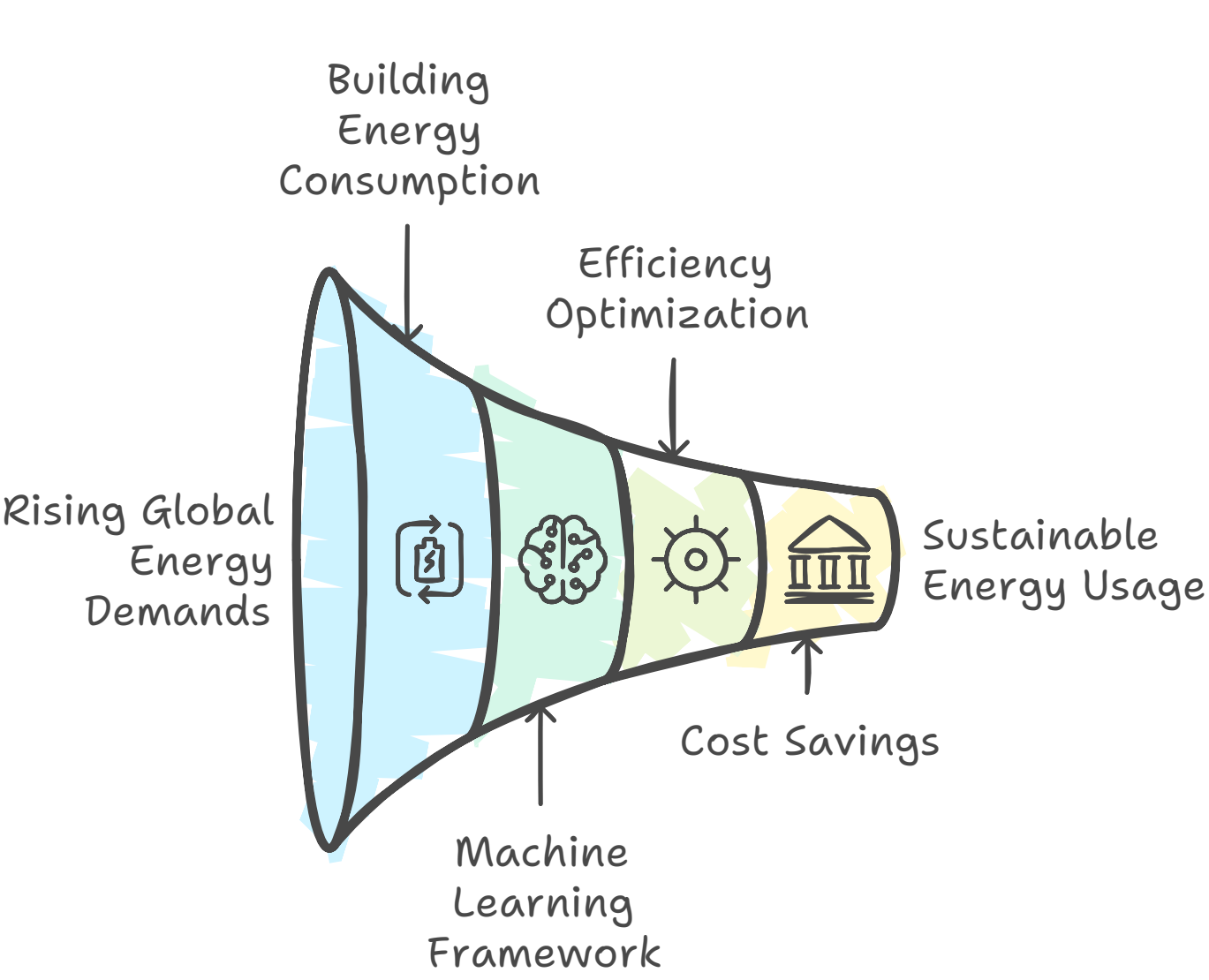
## Project Overview

- **Problem Context**
- Global energy demands are rising, with buildings consuming over 30% of global energy and contributing significantly to $CO_2$ emissions.
- **Objective**
- Develop a machine learning-based framework to accurately predict energy consumption in low-energy buildings, optimizing for efficiency and cost savings.
- **Impact:**
- By understanding and predicting energy needs, we aim to reduce waste, lower operational costs, and support sustainable energy usage.
- **Audience Relevance:**
- Building owners, facility managers, and sustainability advocates can leverage this framework for smarter, data-driven energy management.

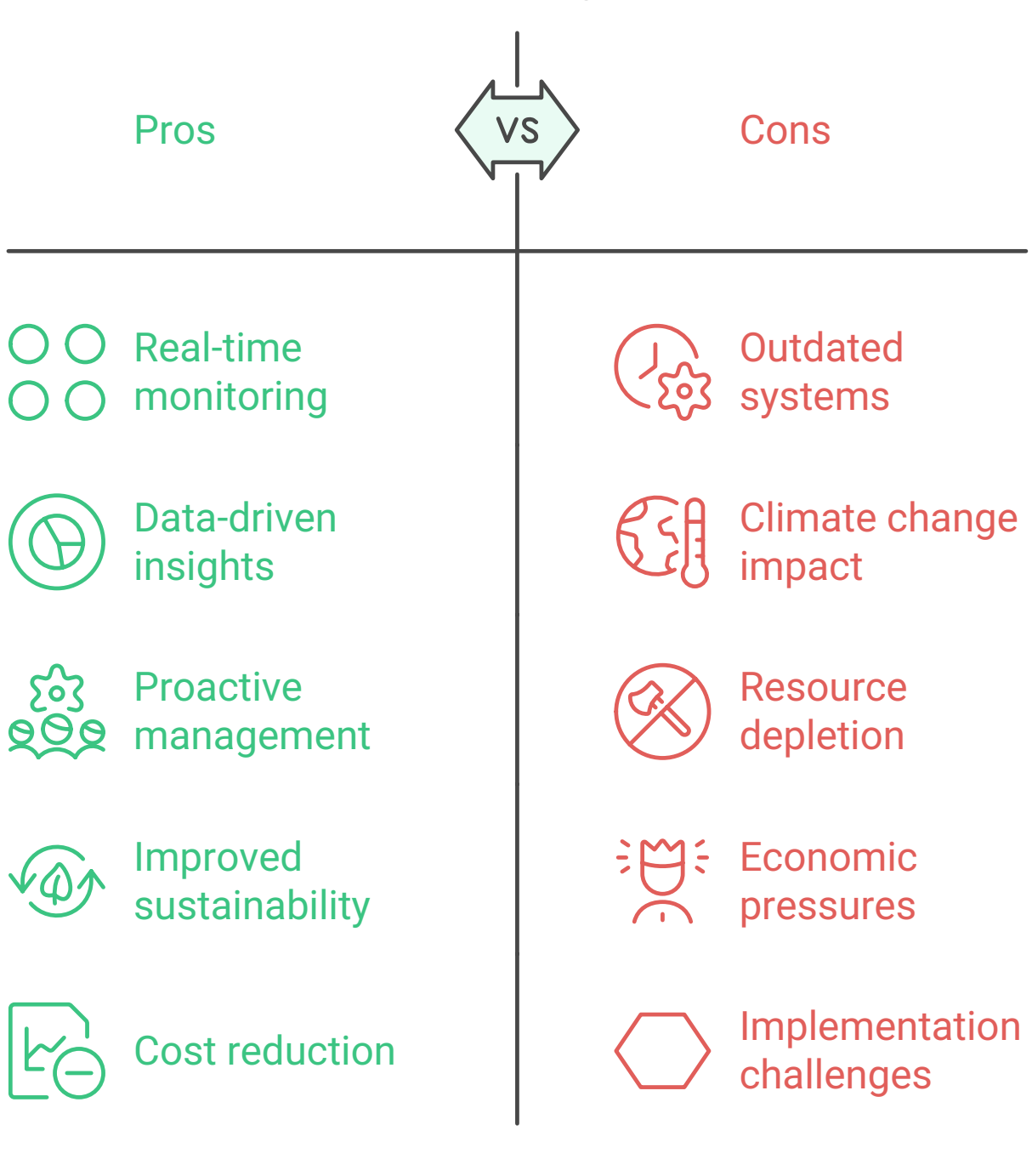
### Energy Management Optimization Funnel


Building Energy Consumption
Efficiency Optimization
Rising Global Energy Demands
Sustainable Energy Usage
Cost Savings
Machine Learning Framework

## Introduction

- **The Energy Crisis**
- Inefficient energy usage in buildings leads to wastage, higher costs, and negative environmental impact. This is aggravated by outdated management systems that don't account for dynamic conditions like occupancy and weather changes.
- **Contemporary Issues**
- Climate change, resource depletion, and economic pressures are pushing for energy optimization.
- **Role of Smart Tech**
- IoT and machine learning enable real-time monitoring, data-driven insights, and proactive energy management.
- **Project's Unique Angle**
- Focus on low-energy buildings where energy prediction and optimization can make a substantial difference in performance and sustainability.

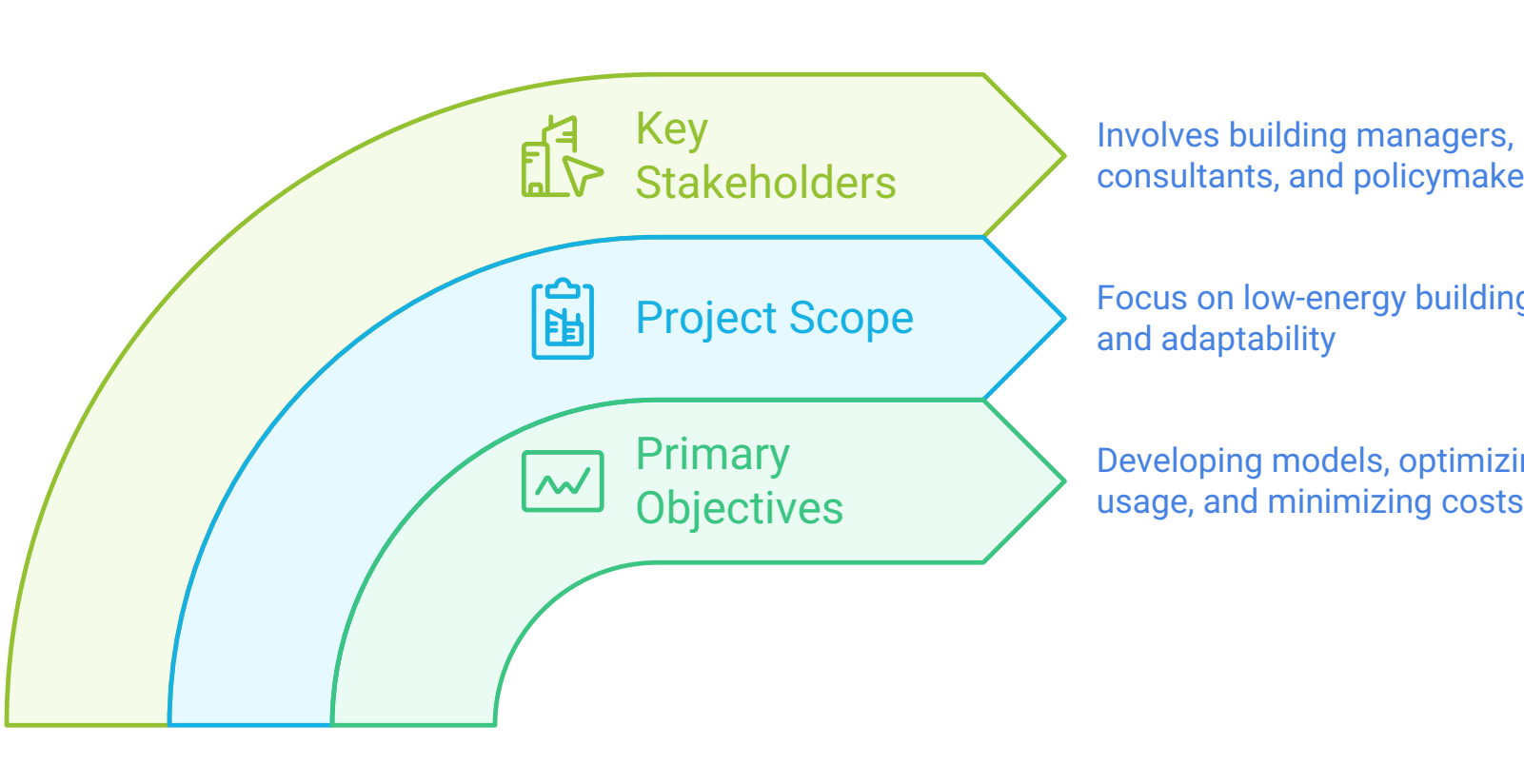
### Smart Tech in Energy Management


Pros
VS
Cons
Real-time monitoring
Data-driven insights
Proactive management
Improved sustainability
Cost reduction
Outdated systems
Climate change impact
Resource depletion
Economic pressures
Implementation challenges

## Objectives and Scope

- **Primary Objectives:**
  - Develop predictive models for energy consumption using machine learning.
  - Optimize energy usage by providing actionable insights to building managers.
  - Minimize operational costs through data-driven decisions.
- **Project Scope:**
  - Targeted at low-energy buildings with specific needs and dynamic variables affecting energy use.
  - Framework adaptable to different building types, with potential applications in both residential and commercial contexts.
- **Key Stakeholders:** Building managers, energy consultants, policymakers aiming to reduce environmental impact.

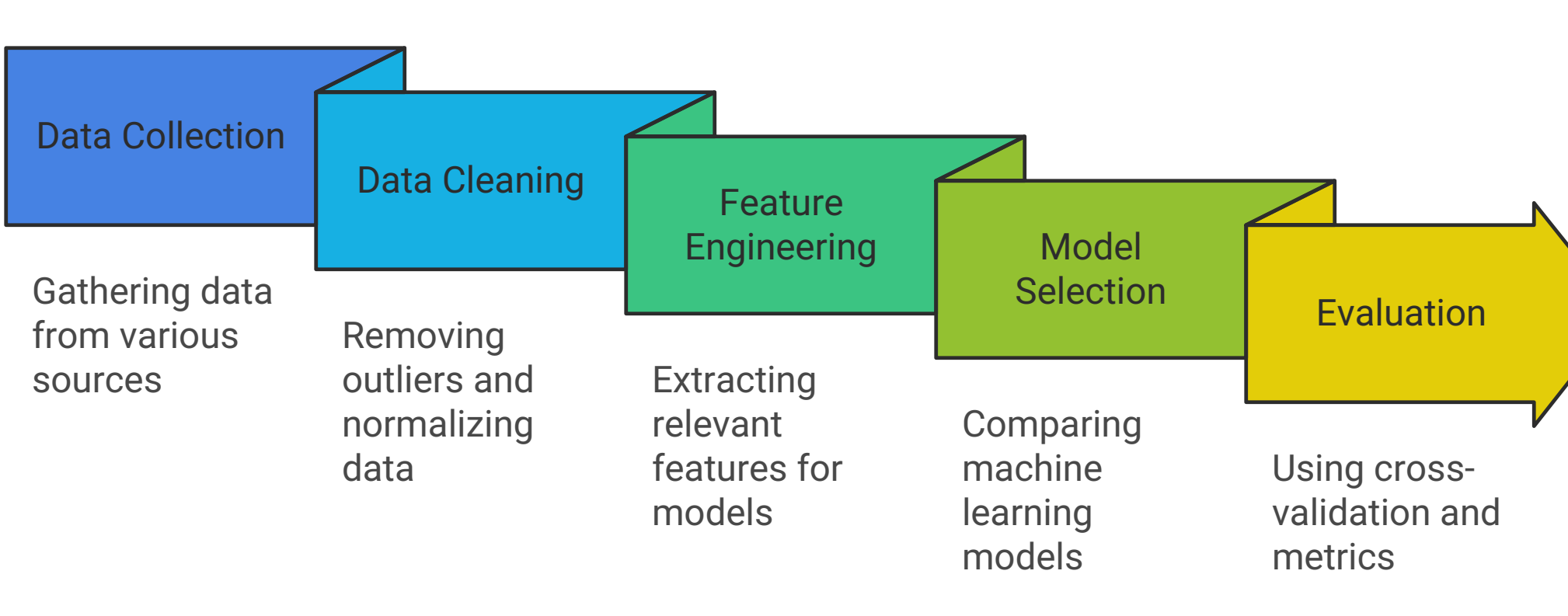
### Energy Optimization Project Overview


Key Stakeholders
Involves building managers, consultants, and policymakers
Project Scope
Focus on low-energy buildings and adaptability
Primary Objectives
Developing models, optimizing usage, and minimizing costs

## Methodology

- **Data Collection:**
  - Sources: Smart meters, environmental sensors, building management systems, and weather data.
  - Collected data includes energy use, occupancy, temperature, humidity, and external weather conditions.
- **Data Processing Techniques:**
  - **Cleaning:** Removal of outliers, normalization for consistency, and handling missing values.
  - **Feature Engineering:** Extracting relevant time-based and categorical features for better model input (e.g., day, month, occupancy patterns).
- **Model Selection:** Compared various machine learning models for accuracy, efficiency, and scalability in low-energy buildings.
- **Evaluation:** K-fold cross-validation and performance metrics like RMSE and R² for objective comparison.

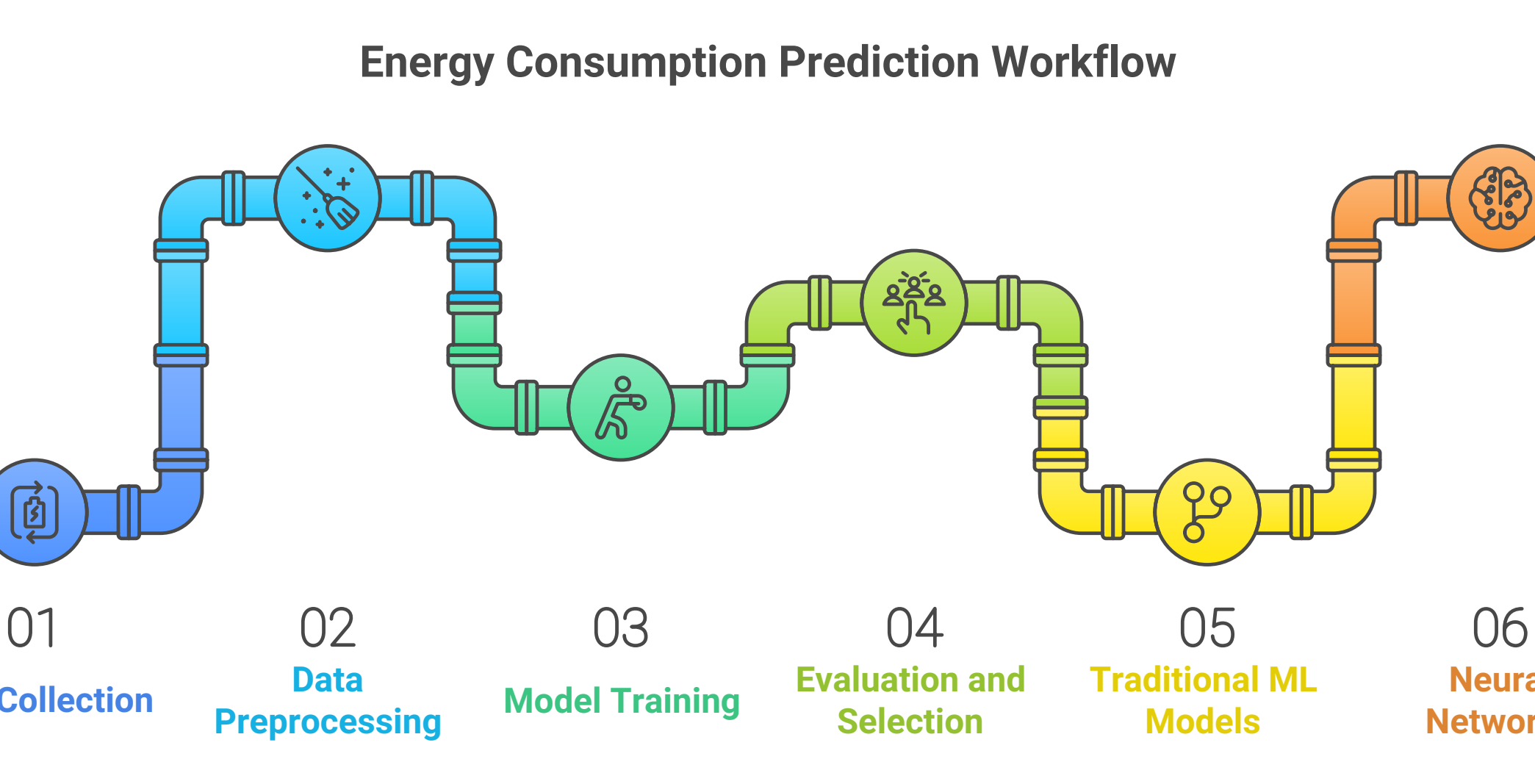
### Energy Consumption Prediction Process


Data Collection
Gathering data from various sources
Data Cleaning
Removing outliers and normalizing data
Feature Engineering
Extracting relevant features for models
Model Selection
Comparing machine learning models
Evaluation
Using cross-validation and metrics

## Design Flow

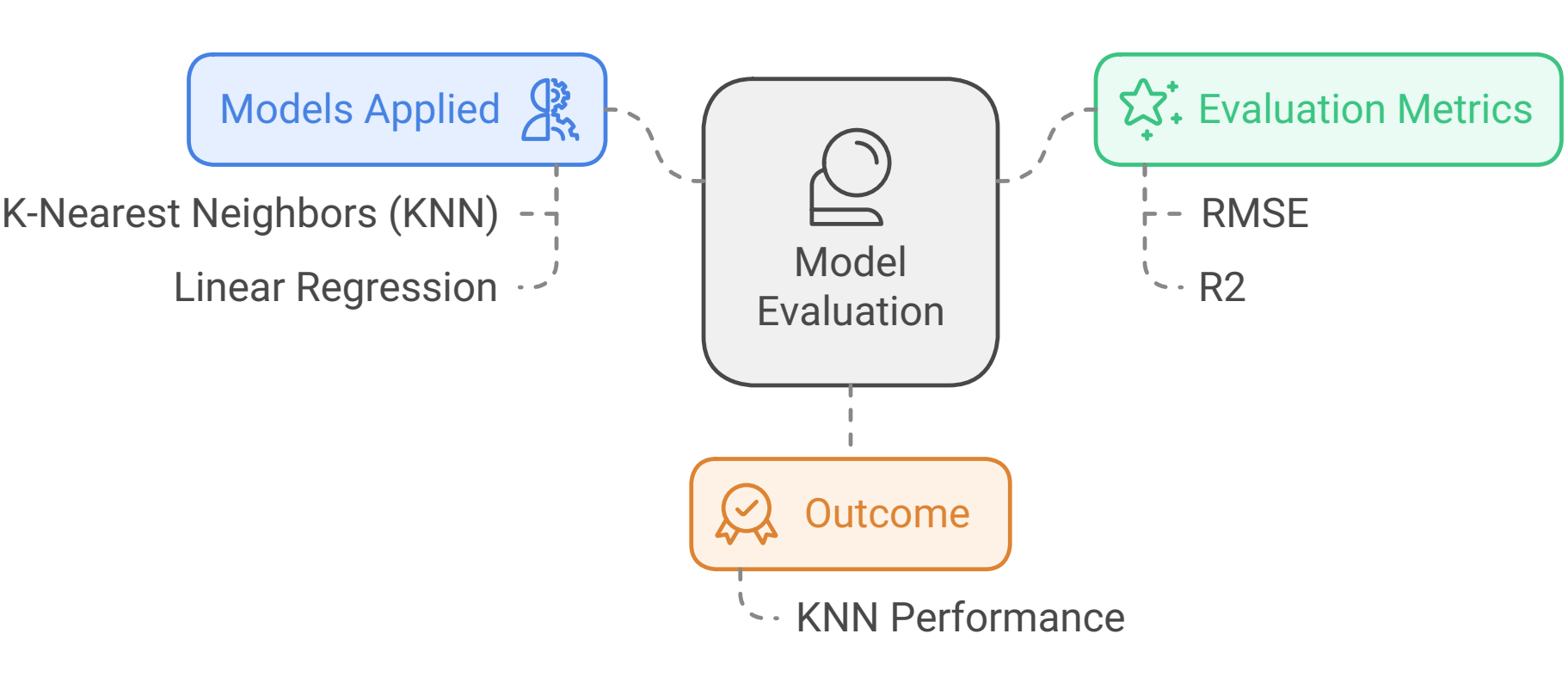
- **Design Phases:**
  - **Data Collection:** Collecting comprehensive multivariate and time-series data on building energy use.
  - **Data Preprocessing:** Cleaning, normalization, and one-hot encoding for categorical data.
  - **Model Training:** Testing several ML models on the preprocessed data.
  - **Evaluation and Selection:** Based on performance metrics, select the most effective model.
- **Approach Types:**
  - **Traditional ML Models:** Linear Regression, Ridge, and Lasso for structured prediction.
  - **Neural Networks:** Potential for future, more complex data patterns and adaptability.
- **Workflow Diagram:** Simple flowchart of the design from data to evaluation and deployment.

### Energy Consumption Prediction Workflow


01
Data Collection
02
Data Preprocessing
03
Model Training
04
Evaluation and Selection
05
Traditional ML Models
06
Neural Networks

## Model Development

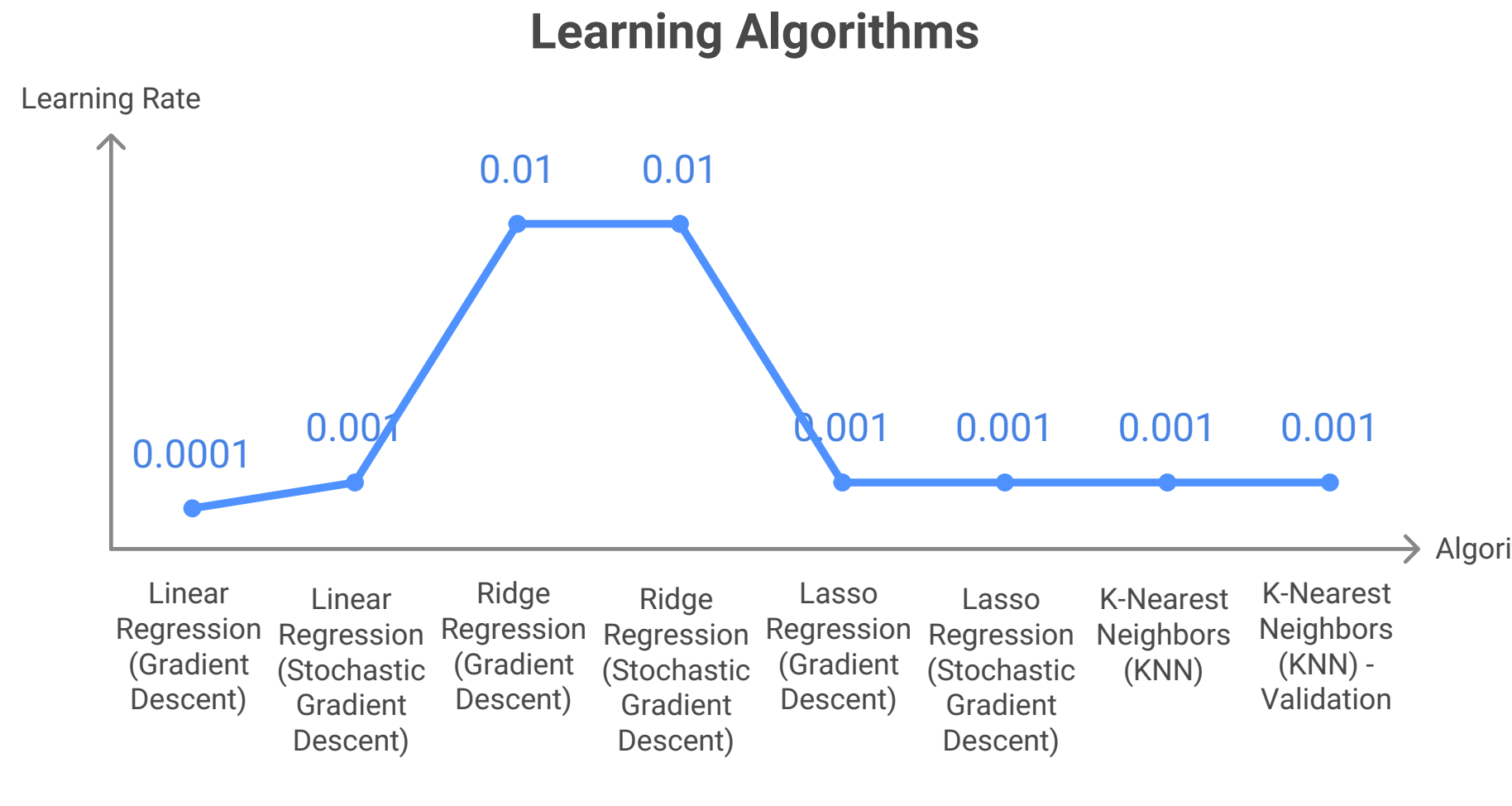
- **Models Applied:**
  - **K-Nearest Neighbors (KNN):** Effective for spatial pattern recognition.
  - **Linear Regression & Variants:** Ridge and Lasso were tested to manage overfitting and multicollinearity.
- **Evaluation Metrics:**
  - **RMSE:** Measures average prediction error, ideal for assessing accuracy.
  - **R²:** Indicates model fit quality, helping gauge overall effectiveness.
- **Outcome:** KNN performed best with an RMSE of 71.69 and R² of 0.54, indicating a solid balance of accuracy and predictability for energy needs.


Models Applied
K-Nearest Neighbors (KNN)
Linear Regression
Model Evaluation
Evaluation Metrics
RMSE
R2
Outcome
KNN Performance

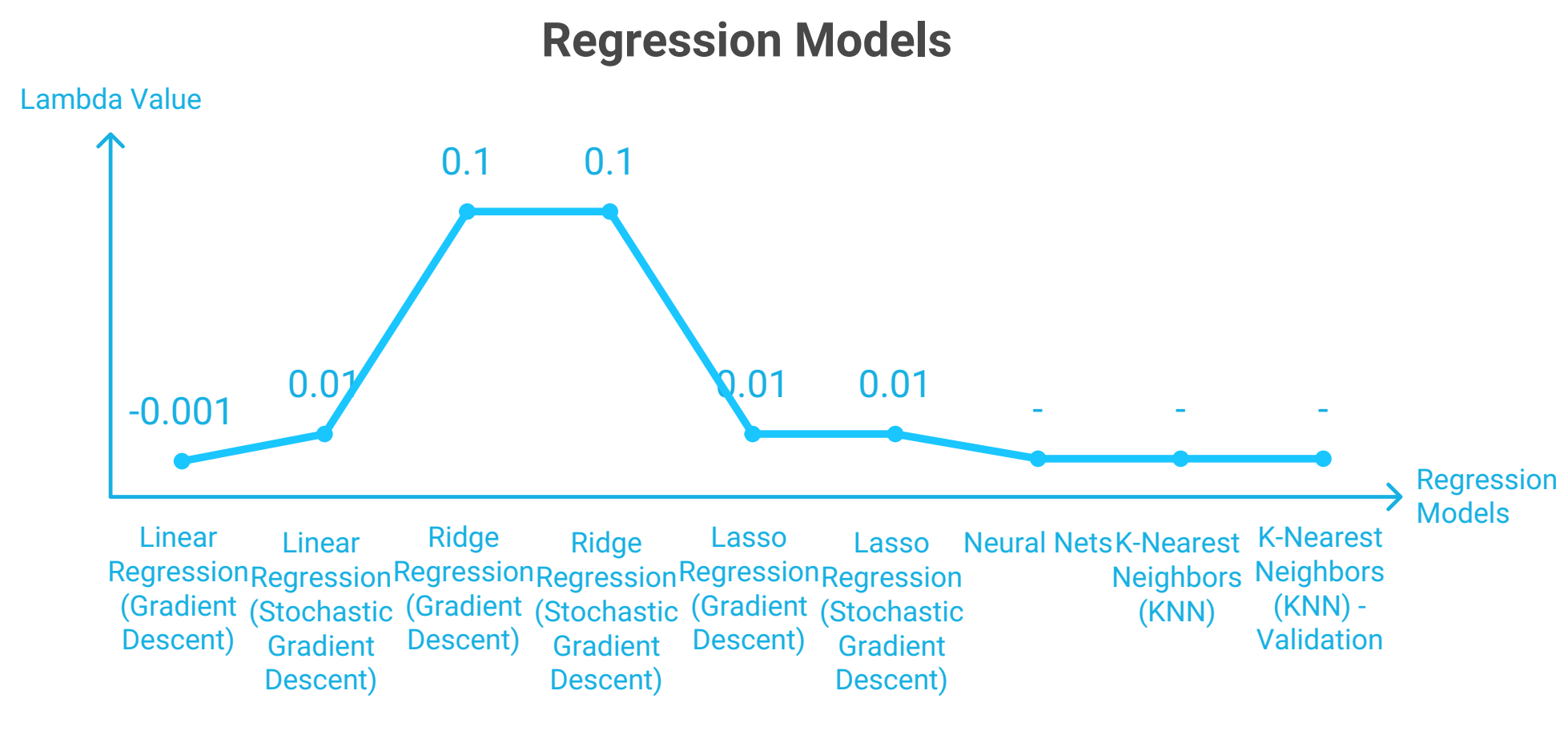
## Results and Analysis

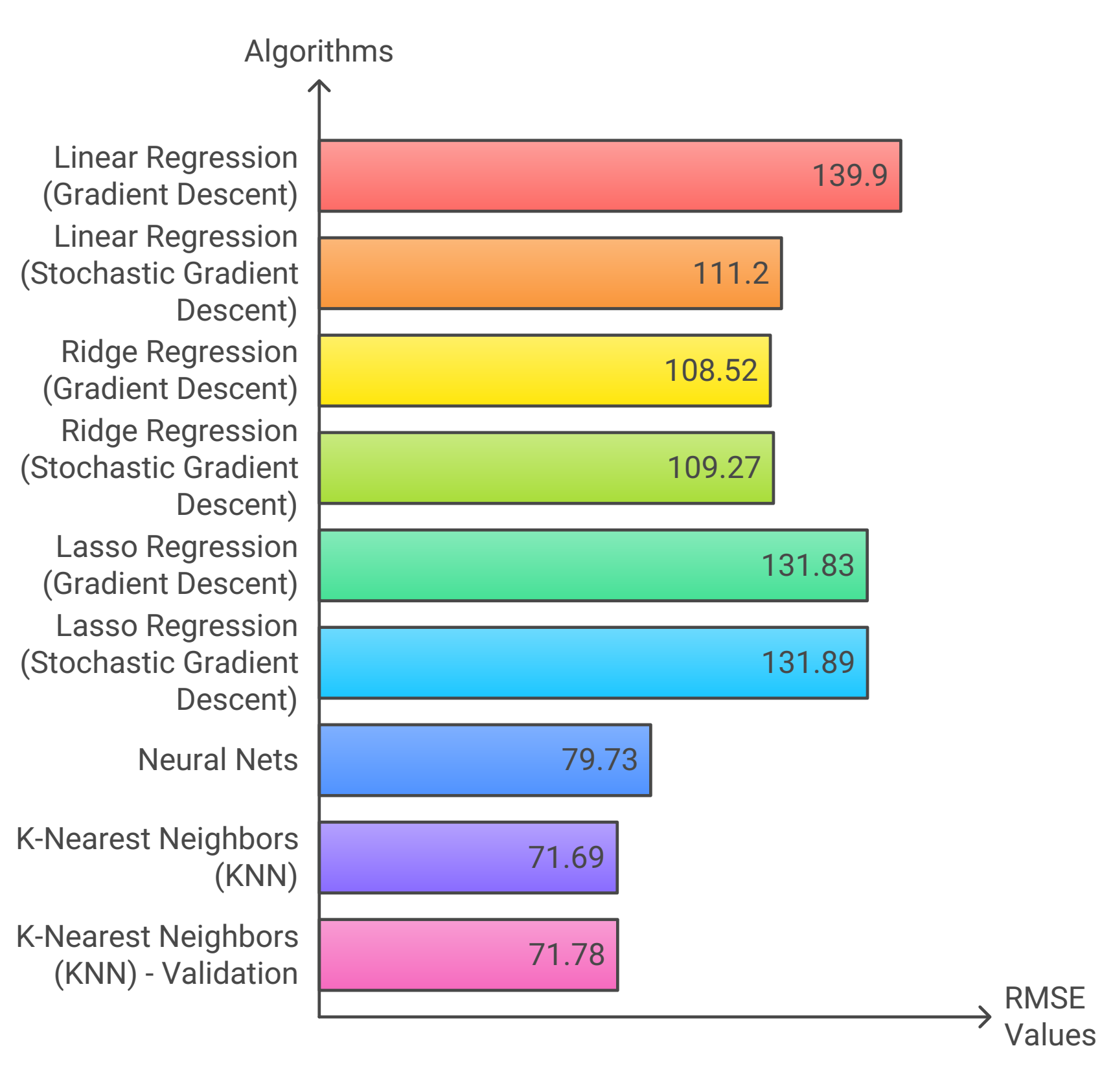
- **Best-Performing Model:** KNN outperformed other models due to its ability to handle complex relationships between data points.
- **Results Summary:**
  - **KNN:** RMSE = 71.69, R² = 0.54 – best accuracy for prediction tasks.
  - **Linear Regression Models:** Ridge and Lasso provided decent alternatives but had lower R² scores.
- **Visuals:** Graphs showing RMSE and R² comparisons across models.
- **Key Insight:** Machine learning models, especially KNN, are promising for energy prediction, though each model's suitability depends on specific data and building characteristics.

### Learning Rates of Machine Learning Algorithms

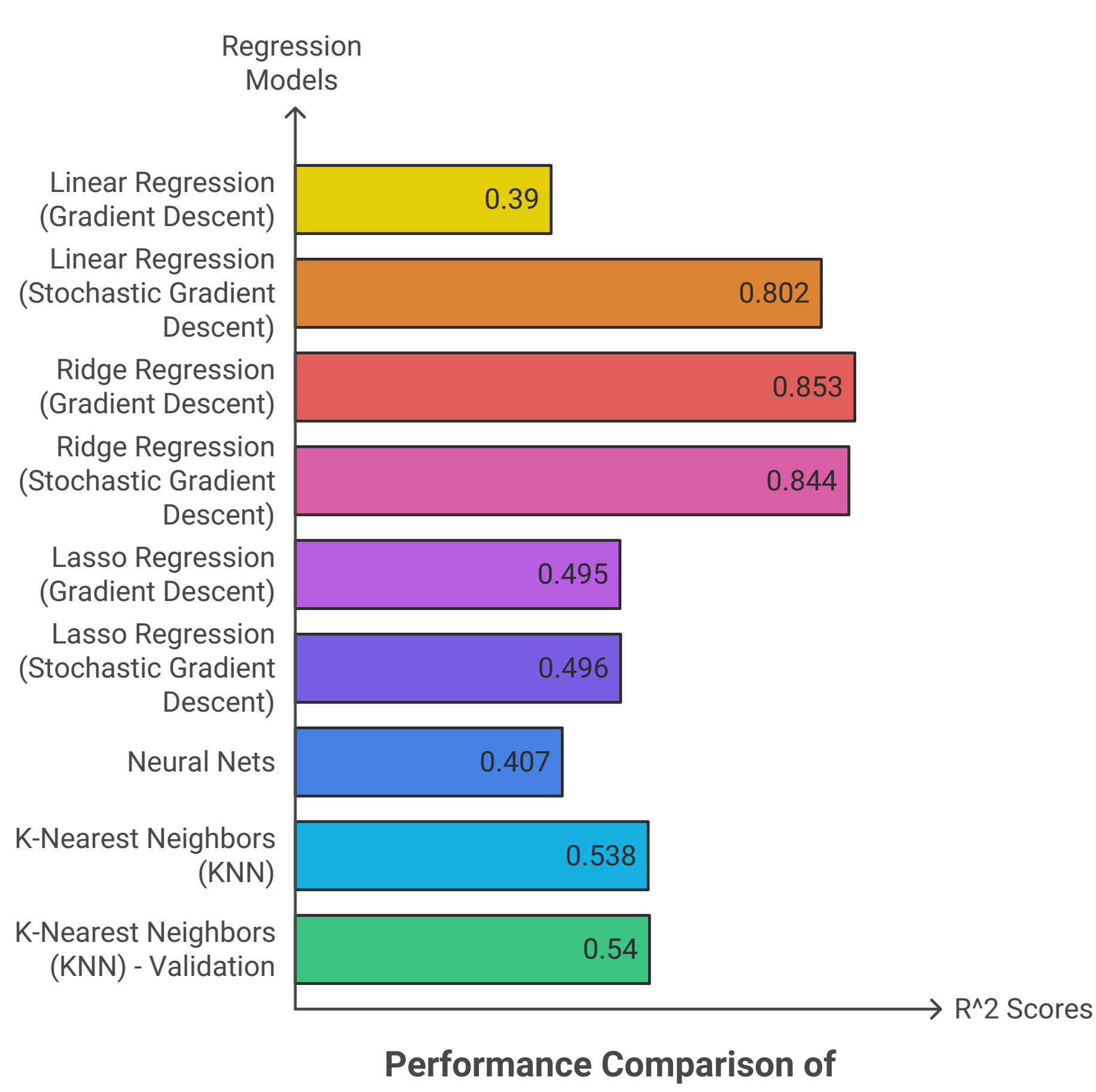


### Lambda Values for Different Regression Models



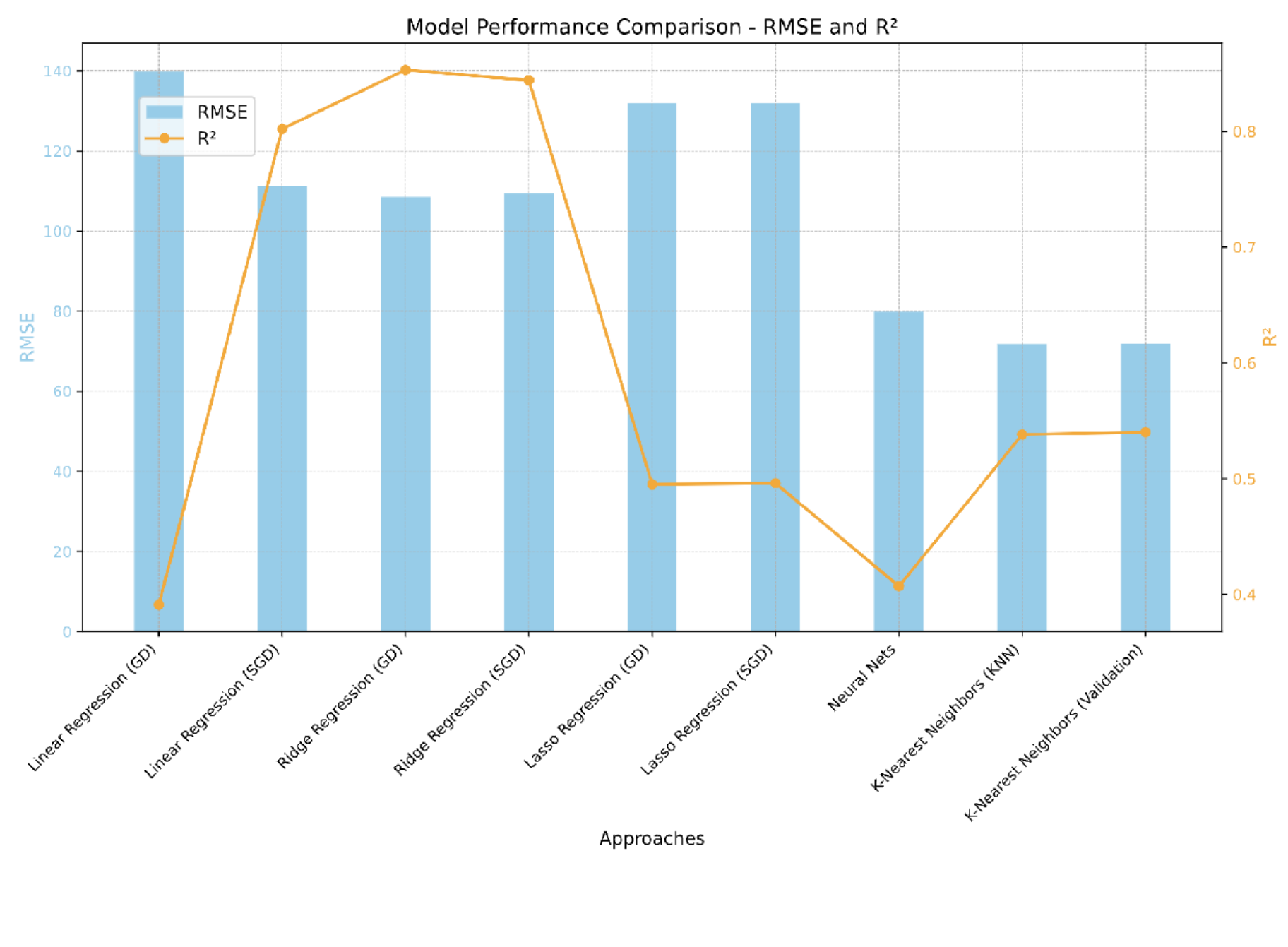


### RMSE Comparison of Machine Learning Algorithms



### Performance Comparison of Regression Models

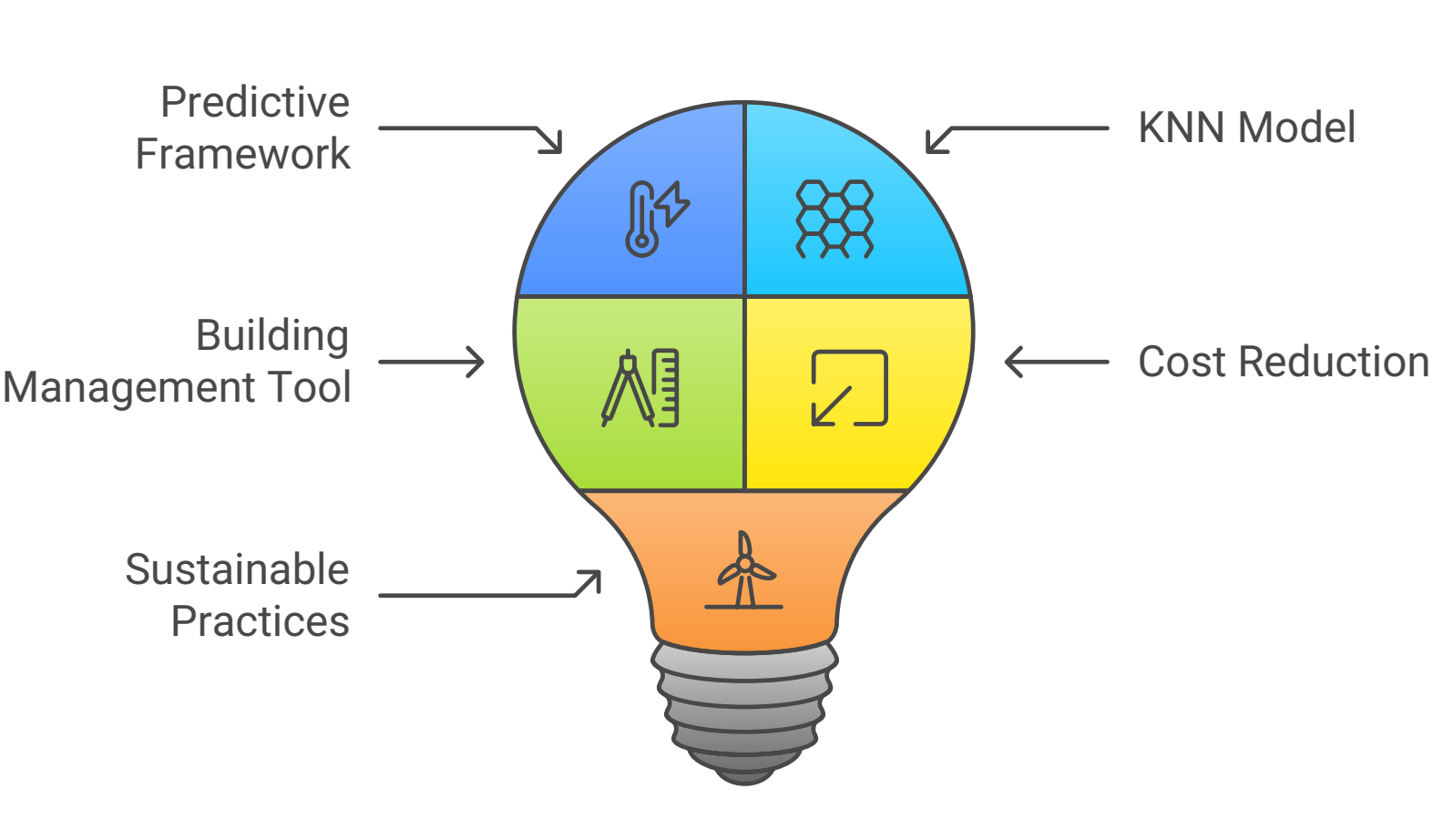
### Difference of the RMSE and R²



## Conclusion

- **Summary of Findings:**
  - Successfully developed a predictive framework for energy consumption, particularly suited for low-energy buildings.
  - The KNN model stood out as an effective choice, balancing predictive accuracy and model efficiency.
- **Impact of Project:**
  - Provides building managers with a reliable tool to forecast and manage energy usage.
  - Supports operational cost reductions and promotes sustainable energy management practices.
- **Core Takeaways:** Machine learning can significantly enhance energy management by predicting needs and allowing for real-time optimization.

### Enhancing Energy Management with AI


Predictive Framework
Building Management Tool
Sustainable Practices
KNN Model
Cost Reduction

# Thank You